Buffers, their Mechanism of Action & Biological Buffers

Lecture Objective

At the end of the lecture, students will be able to understand the following learning objectives;

1. Buffer, and its preparation.
2. The mechanism of action of buffers.
3. Biological buffers and their importance.

BUFFERS

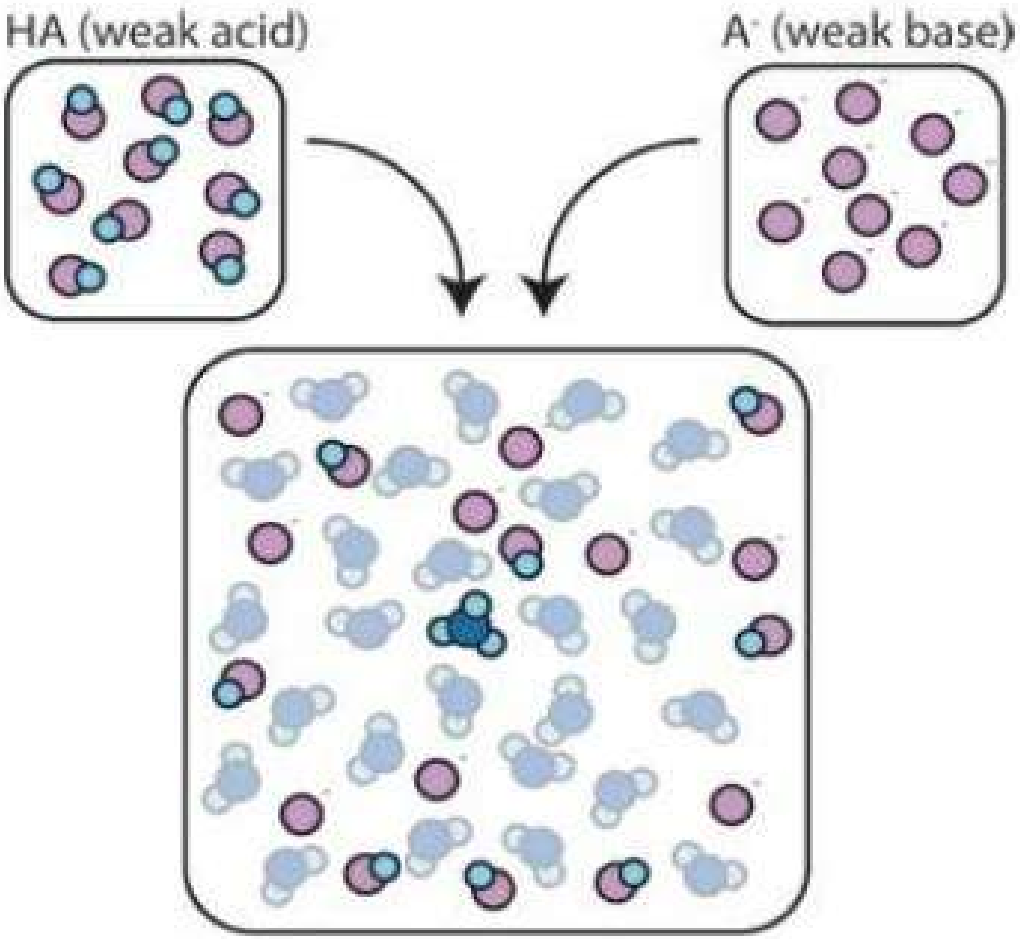
- A solution which resists any change of pH when a small amount of a strong acid or a strong base is added.
- ***Buffers in solution:*** An acid-base conjugate pair can act as a buffer, resisting changes in pH.

OR

- A aqueous solution mixture made in weak base with conjugate acid or weak acid with conjugate base.
- ***Weak acids*** are only partly ionized in solution.
- ***Weak base*** is a chemical base that does not ionize fully in an aqueous solution.

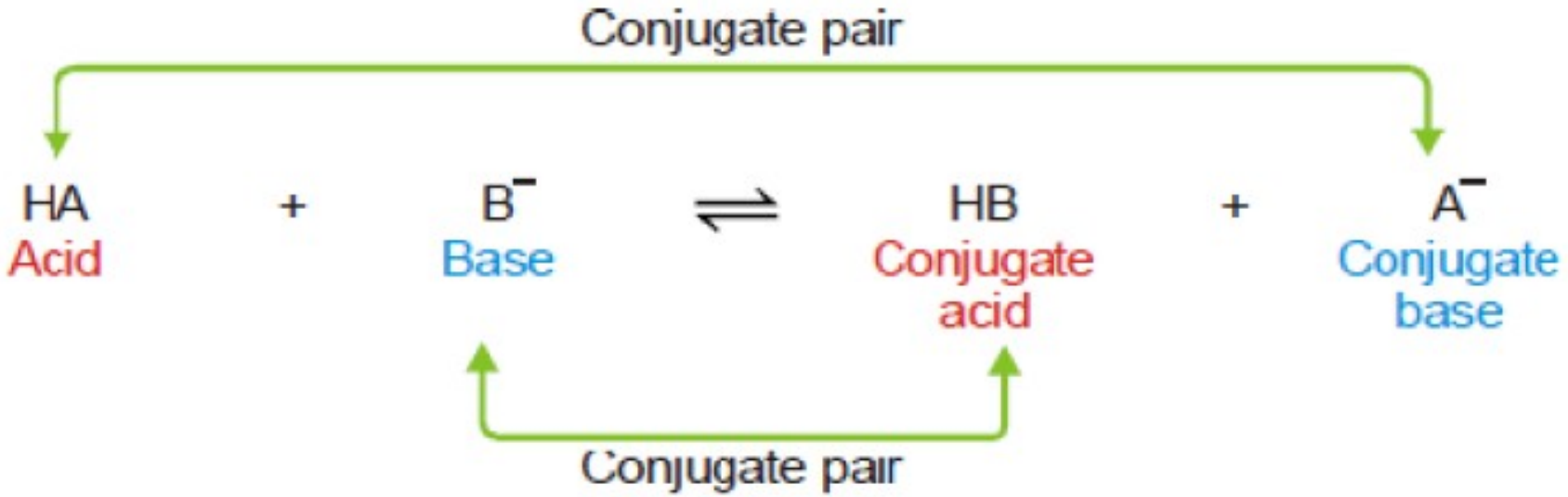
Buffer Solutions

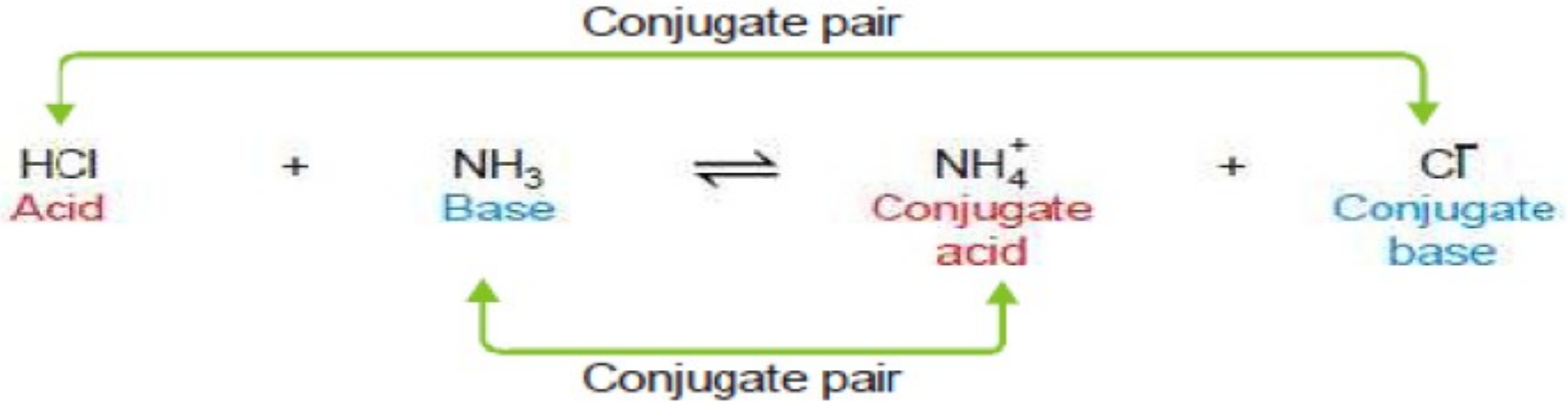
- DEFINITION: A buffer solution contains a weak acid mixed with its conjugate base (or weak base and conjugate acid)
- Buffers resist changes in pH when a **small** amount of a strong acid or base is added to it.



$$HA \prod H^+ + A^-$$

Conjugate Acid-Base Pairs





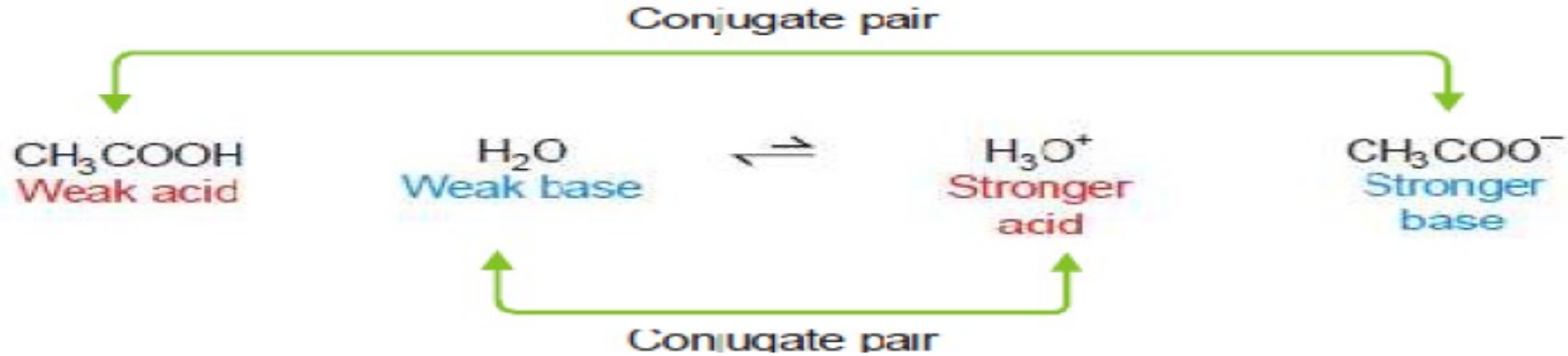
Types of Buffers

- According to the ionic strength of a solution, there are usually two common types of Buffers;

1. Acid Buffers
2. Basic Buffers

Acid Buffers

- Solution mixture containing acid and salt of same acid with strong base.
- Acidic buffer solutions are commonly made from a ***weak acid*** and ***one of its salts*** **-** often a sodium salt.
- An acidic buffer solution is simply one which has a pH ***less than 7.***

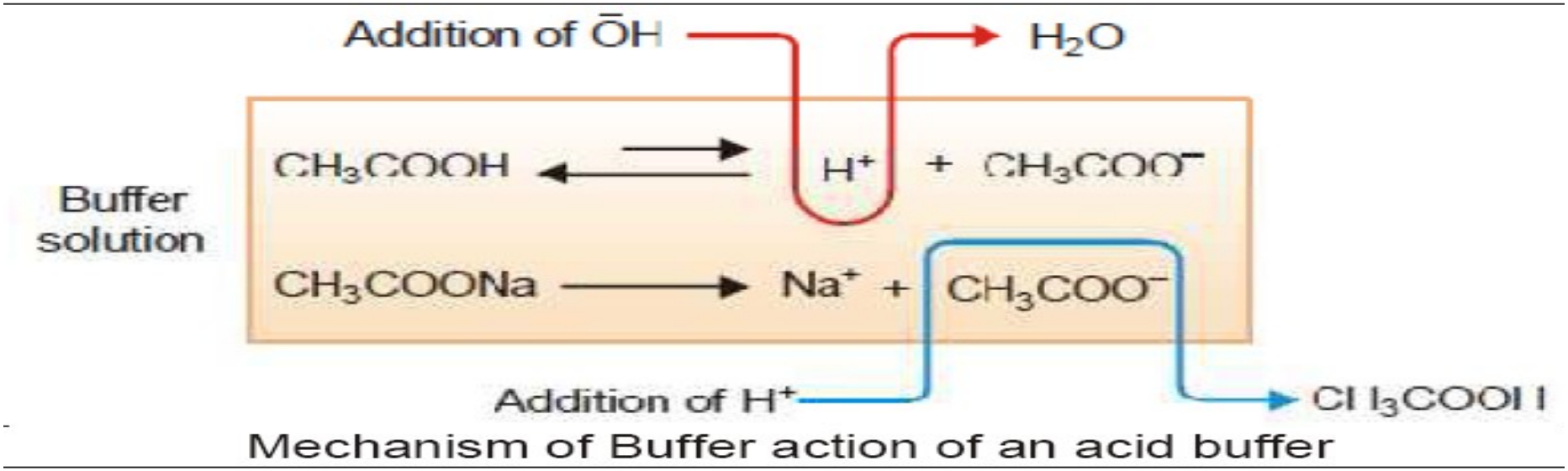


Example

- The examples of acid buffers include; $CH_3COOH + CH_3COONa$.
- ***Increase of (addition of HCl) H^+ ions*** - form unionized CH_3COOH.
- ***Increase of OH^- ions*** – combine with H^+ ions – forms Water (H_2O) molecule

$$CH_3COOH \rightleftharpoons H^+ + CH_3COO^-$$

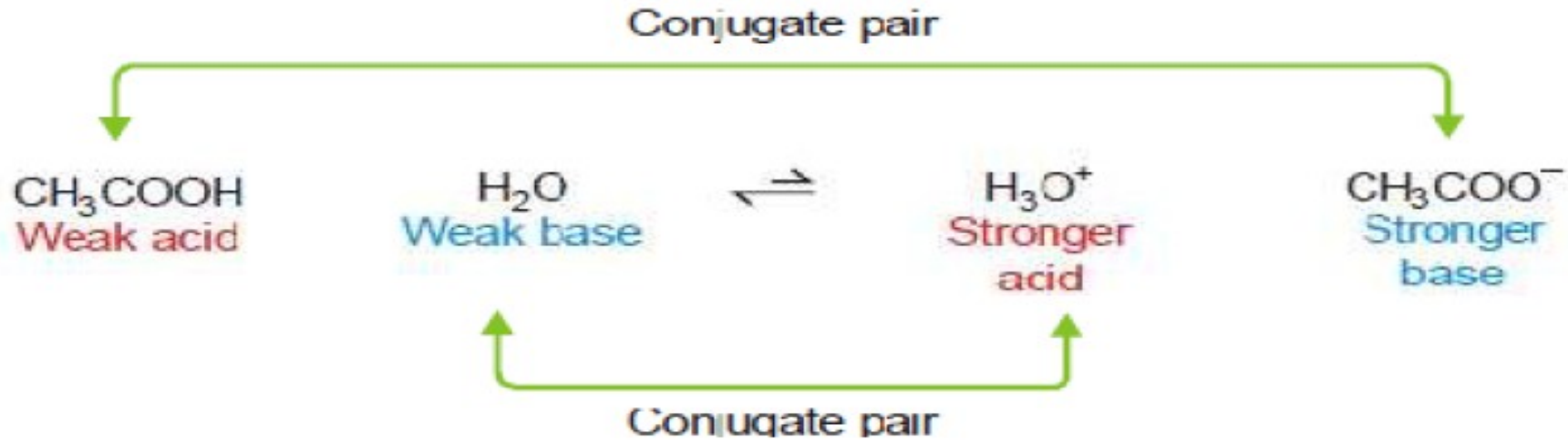
$$CH_3COONa \longrightarrow Na^+ + CH_3COO^-$$



Mechanism of Buffer action of an acid buffer

Basic Buffers

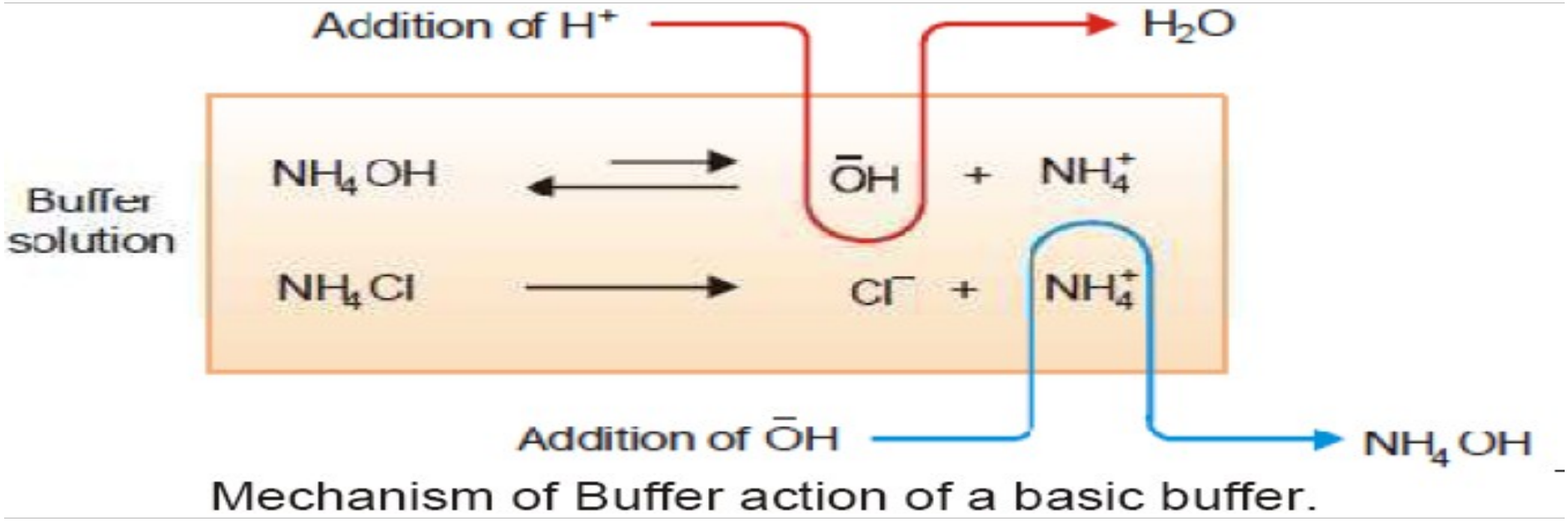
- Solution mixture containing base and salt of the same base with strong acid.
- Alkaline buffer solutions are commonly made from a ***weak base*** and ***one of its salts***.
- An alkaline buffer solution has a pH greater than 7.



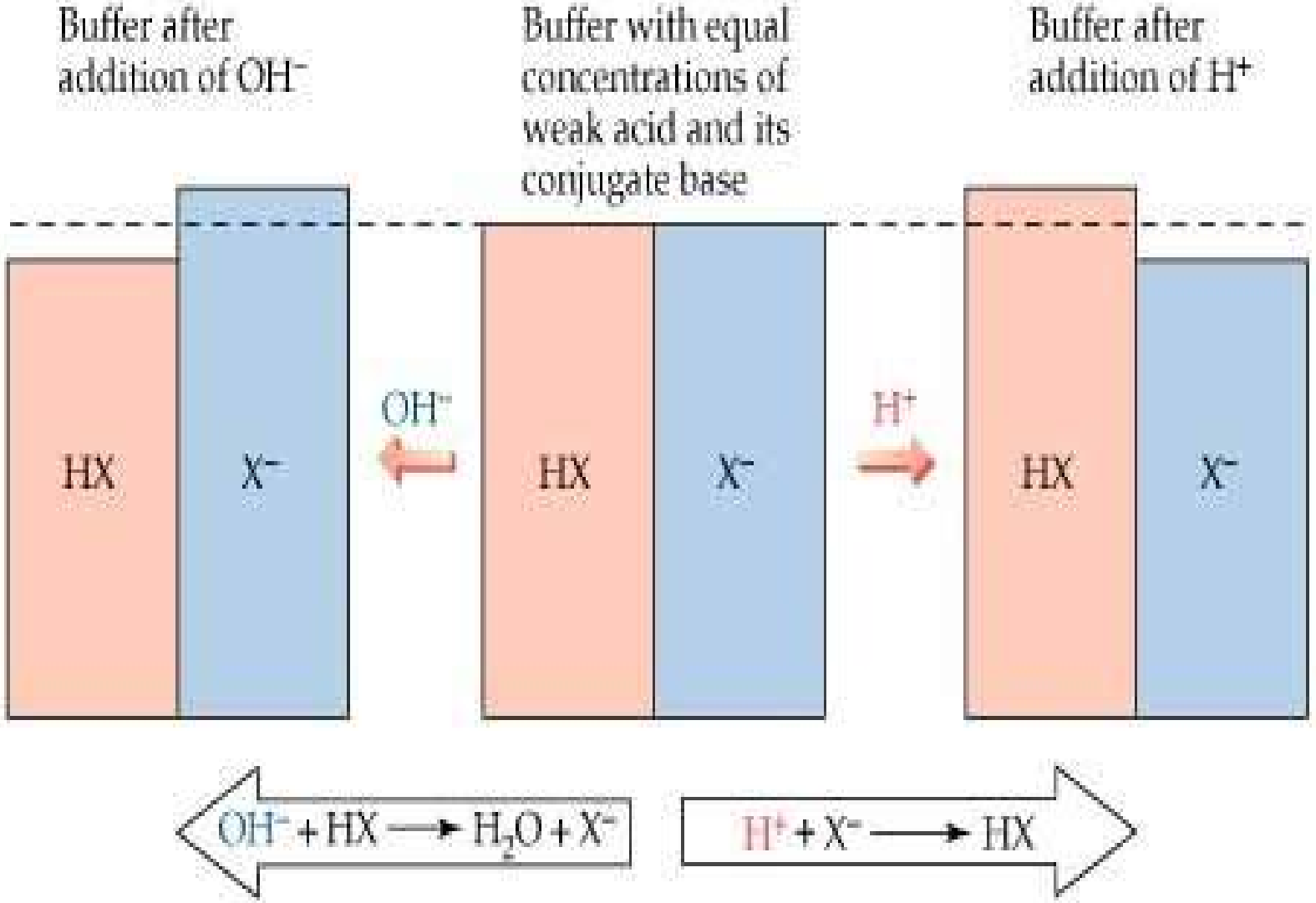
Example

- The examples of basic buffers include; NH_4OH + NH_4Cl.
- ***Increase of H^+ ions*** - form H_2O (all H^+ neutralized- Original buffer pH restored).
- ***Increase of OH^- ions*** – associate with excess NH_4+ ions to form unassociated NH_4OH

$$NH_4OH \rightleftharpoons NH_4^+ + OH^-$$



Mechanism of Buffer action of a basic buffer.


Buffer after addition of OH^-
Buffer with equal concentrations of weak acid and its conjugate base
Buffer after addition of H^+
HX
X^-
OH^-
HX
X^-
H^+
HX
X^-
$OH^- + HX \longrightarrow H_2O + X^-$
$H^+ + X^- \longrightarrow HX$

Henderson-Hasselbalch Equation

- In case of strong acids and strong bases, the calculation of the ionization constants was easy.
- For weak bases and weak acids, the calculations are rather difficult to be made as weak acids or weak bases hardly ionize.
- Henderson-Hasselbalch equation is generally used for such predictions.

Key to understanding what the pH of this solution would be---is the **Henderson-Hasselbalch Eqn.**

$$K_a = \frac{[H^+][A^-]}{[HA]}$$

take logarithm of both sides:

$$\log K_a = \log\left(\frac{[H^+][A^-]}{[HA]}\right) = \log[H^+] + \log\frac{[A^-]}{[HA]}$$

swap--$\log K_a$ and log[H+] to opposite sides of eqn:

$$-\log K_a = -\log[H^+] + \log\frac{[HA]}{[A^-]}$$

$$pK_a = pH + \log\frac{[A^-]}{[HA]} \quad \longleftarrow \quad \textbf{HH-eqn.}$$

Body Buffers

- Buffers cannot remove H+ ions from the body.
- It temporarily act as absorbent to reduce the free H+ ions concentration.

Body contain three (03) buffer system;

1. Bicarbonate buffer
2. Phosphate buffer
3. Protein buffer

Bicarbonate Buffers

- Some of CO_2 (the end product of cellular metabolism) is carried to lungs for elimination.
- Remaining CO_2 is dissolves in body fluids, forming Carbonic acid, that further dissociates into bicarbonate (HCO^-) and hydronium (H_3O^+) ions.
- More HCO_3- is supplied by kidneys;

$$CO_2 + H_2O \rightleftharpoons H_2CO_3$$

$$H_2CO_3 + H_2O \rightleftharpoons H_3O + HCO_3^-$$

$$H_2CO_3 \rightleftharpoons H^+ + HCO_3^-$$

Henderson-Hasselbalch Equation

For Bicarbonate Buffer

$$pH = pKa + \log \frac{[HCO_3^-]}{[H_2CO_3]}$$

Exercise: The dissociation constant for $H_2CO_3^-$ = 6.1, molar concentration of bicarbonate is 0.024 mol/L, and carbonic acid is 0.0012 mol/L. Calculate the pH?

- At pH of blood plasma (7.4) very little H_2CO_3 present in comparison with HCO_3^-. In case when H+ concentration increases (for e.g., heavy exercise, increase lactate) – reaction equilibrium is shift and H_2CO_3 concentration increases.
- It means that dissolves CO_2 in blood increases and thus increases the partial pressure of CO_2 (gas) in the air space of lungs (extra CO_2 is exhaled).
- Conversely, when blood pH is raised (NH3+ produced during protein catabolism) - H+ ions concentration is lowered and causing more H_2CO_3 to dissociate into H^+ and HCO_3^-
- Hence, CO_2 from lungs to dissolve in blood plasma

- Excess Acid

$$H_2CO_3 \rightleftharpoons H^+ + HCO_3^-$$

- Excess Base / NH_3^+

$$H_3CO_3 + OH^- \rightleftharpoons H_2O + HCO_3^-$$

Concentration of Carbonate and Bicarbonate ions decide the pH of blood

Phosphate Buffers

- This buffer is constituted with Sodium di-hydrogen phosphate (NaH_2PO_4) and Disodium hydrogen phosphate (Na_2HPO_4).
- Present in low concentration in plasma.
- The Phosphate buffer system (HPO_4^{2-} / $H_2PO_4^-$) plays a significant role in plasma and erythrocyte.
- Any acid react with mono-hydrogen phosphate to form di-hydrogen phosphate.

$$HPO_4^{2-} + H^+ \rightleftharpoons H_2PO_4^- + H_2O$$

- The base is neutralized by di-hydrogen phosphate as;

  $H_2PO_4^- + OH^- \rightleftharpoons HPO_4^{2-} + H^+$

- Phosphate buffer system maximally effective at pH close to its pKa = 6.86.
- This means that it resists the pH changes in range of 5.9 – 7.9.
- It is also referred as the buffer of biological fluid.
- Most extracellular fluids and most cytoplasmic compartments have a pH in the range of 6.9 - 7.4

Protein Buffers

- The protein present in plasma and hemoglobin constituted the protein buffer in blood.
- The buffering capacity of protein dependent on the pKa of ionize-able group of amino acids.
- Importantly an amino acids COO^- groups (as proton donor).
- Amino acids contains NH_3^+ groups which like ammonium ion (NH_4^+) can donate protons.
- If acid or H^+ appears / comes in blood, H^+ ions can be neutralized by COO^- group.

$$COO^- + H^+ \rightleftharpoons COOH + H_2O$$

- If base or OH- ions appears in blood, it can be neutralized by NH_3^+ group.

$$NH_3^+ + OH^- \rightleftharpoons NH_2 + H_2O$$

- For e.g., the imidazole group of Histidine (pKa = 6.7) is the most effective contributor of protein buffer. NH_3^+ (Basic group) & C=NH (Neutral group).
- Plasma protein accounts for above 2% of total buffering capacity of the plasma.

Importance of Selective Permeable Membrane

Selectively Permeable Membrane

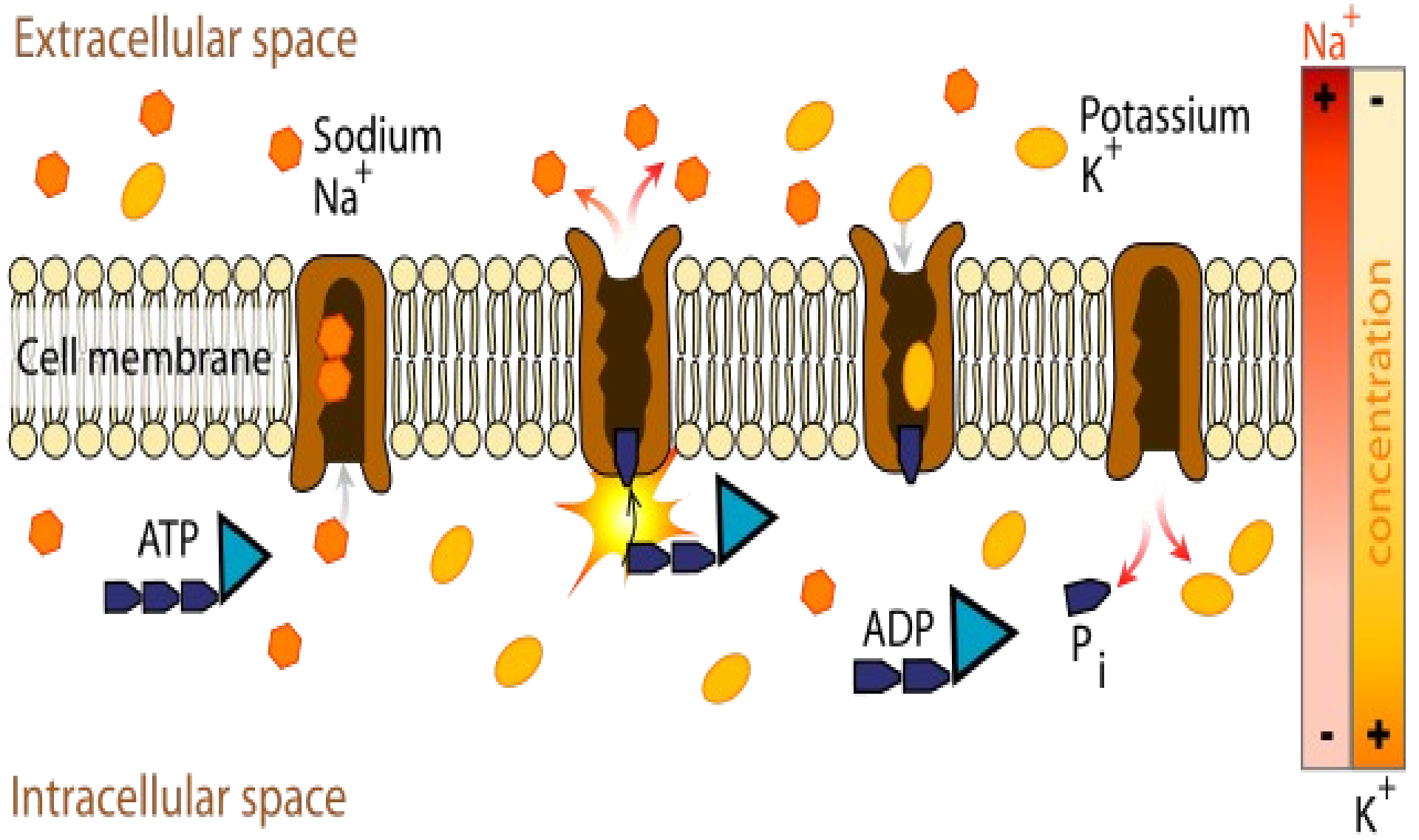
- A **selectively permeable membrane (cell membrane)** is one that permits certain ions or molecules to pass through it by means of active or passive transport.
- ***Active transport:*** require the cell to expend energy to move the materials
- ***Passive transport:*** it can be done without using cellular energy.
- It is important to note that this type of membrane can also block the passage of materials depending on the conditions inside or outside of the cell.

Examples of Selectively Permeable Membrane

- This type of membranes can be found around a variety of cells and places.
- Examples of Selectively Permeable membranes includes the phospholipid bilayer cell membrane that surrounds every cell in our bodies.
- Another example of a selectively permeable membrane is the inner membranes of an egg.

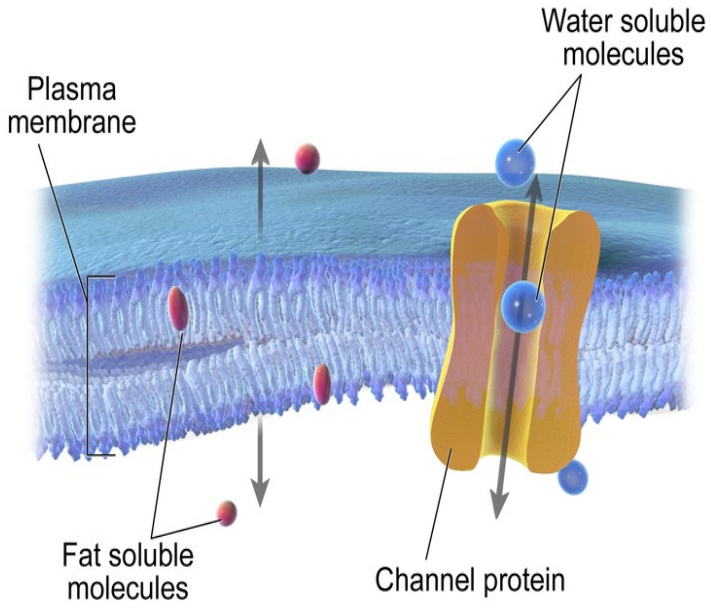
Active transport

- It involves the movement of molecules (across a cell membrane) from a region of their lower concentration to a region of their higher concentration.
- It is usually associated with accumulating high concentrations of molecules that the cell needs, like certain ions, glucose and amino acids.
- The process uses chemical energy, such as adenosine triphosphate (ATP).

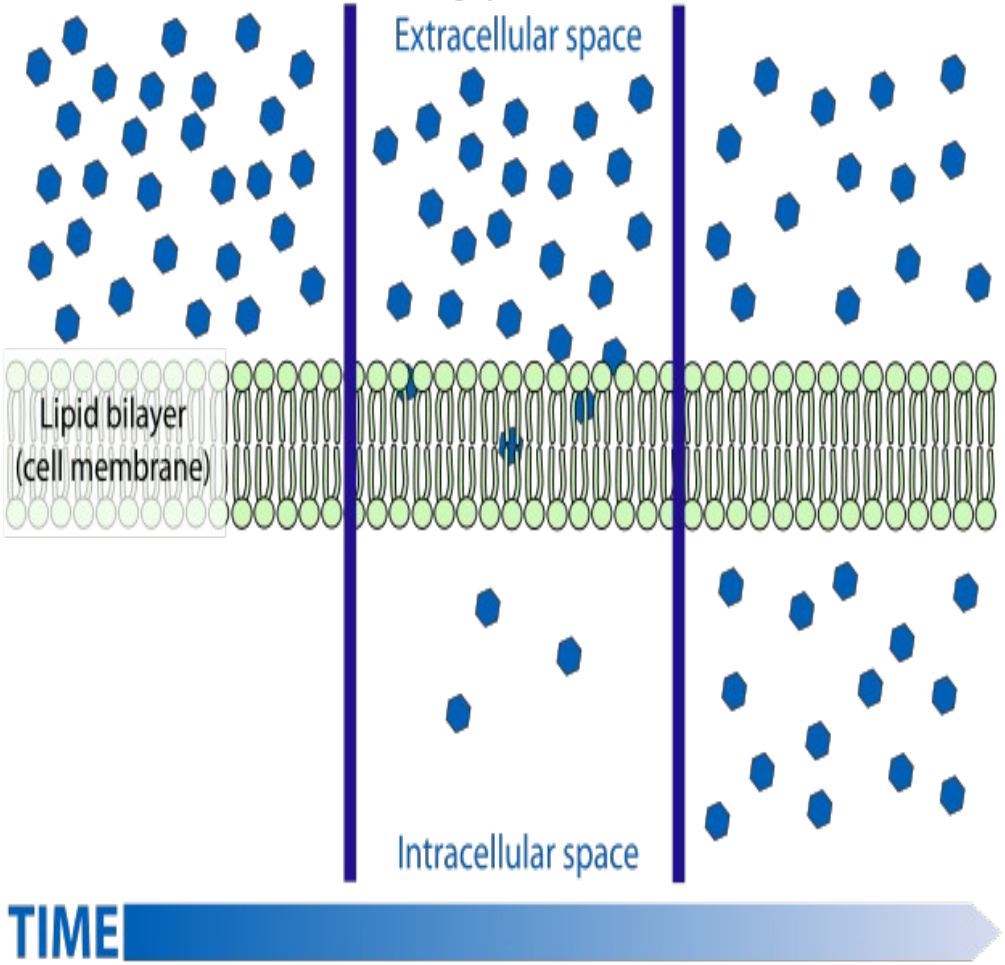

Extracellular space
Sodium
Na+
Potassium
K+
Cell membrane
ATP
ADP
Pi
Na+
+
-
concentration
-
+
K+
Intracellular space

Passive transport

- Passive transport is a movement of biochemical and other atomic or molecular substances across cell membranes without need of energy input.

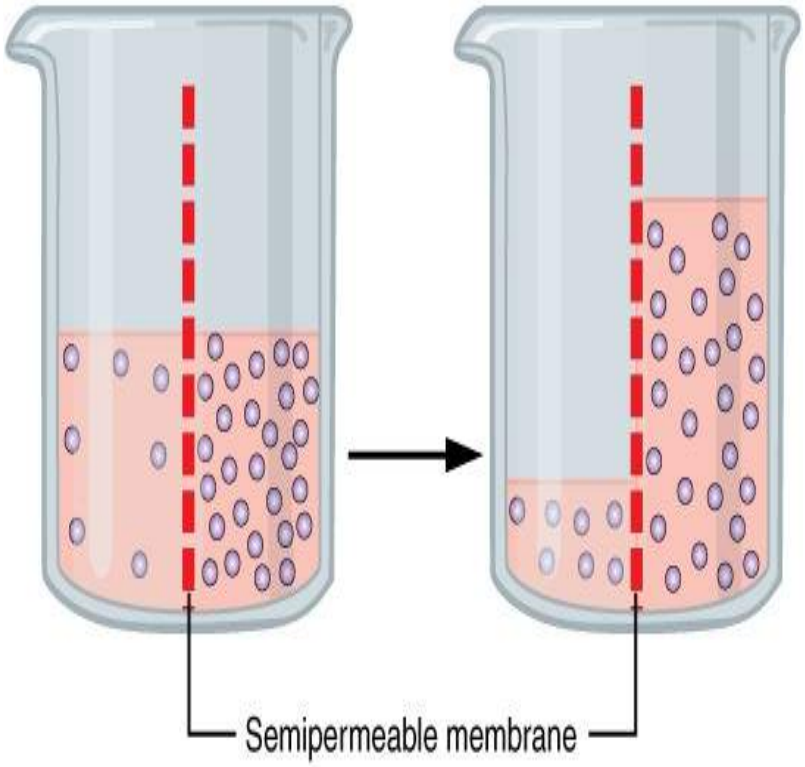


Diffusion Across the Plasma Membrane



Osmosis

- The movement of solvent molecules through a semi-permeable membrane into a region of higher solute concentration, in the direction that tends to equalize the solute concentrations on the two sides.

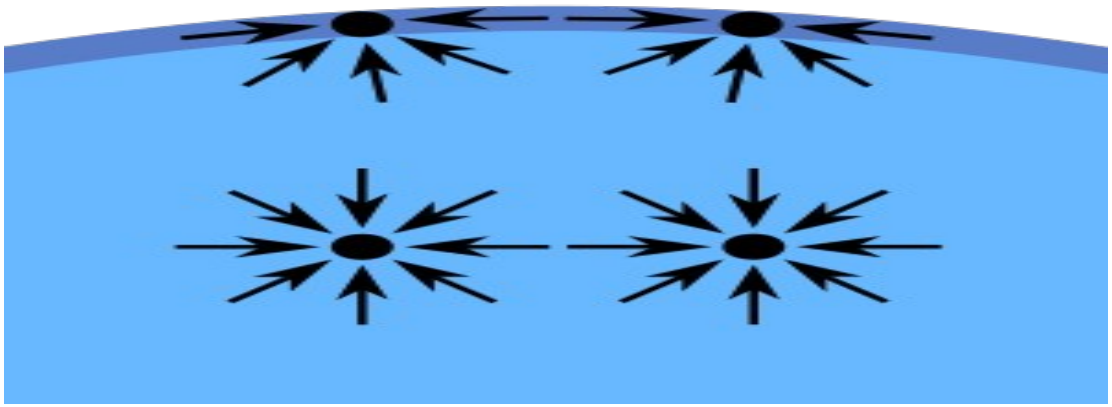


Osmotic Pressure

- Osmotic pressure is the minimum pressure which needs to be applied to a solution to prevent the inward flow of water across a semipermeable membrane.
- In presence of membrane, molecules of water will flow to the area with the highest concentration of solute.
- Pressure created by water moving across a membrane due to osmosis is termed as Osmotic pressure.
- Hence, more the water moving across the membrane, the greater (higher) the osmotic pressure.

Surface Tension

- Surface film of a liquid caused by the attraction of the particles in the surface layer by the bulk of the liquid, which tends to minimize surface area.
- It results from the greater attraction of liquid molecules to each other (due to cohesion) than to the molecules in the air (due to adhesion).
- It permits insects (e.g. water striders), usually denser than water, to float and stride on a water surface.





Viscosity

- Viscosity is a property of the fluid.
- The viscosity of a fluid is a measure of its resistance to gradual deformation by shear stress or tensile stress.
- It opposes the relative motion between the two surfaces of the fluid in a fluid that are moving at different velocities.
- Body fluids also possess this important property. For example blood viscosity is defined as the inherent resistance of blood to flow.

Importance of Viscosity in body fluids

- Increased blood viscosity is the only biological parameter that has been linked with all of the other major cardiovascular risk factors, including high blood pressure, elevated LDL cholesterol, low HDL, type-II diabetes, metabolic syndrome, obesity, smoking, age, and male gender.
- Smoking has been shown to cause blood viscosity to surge by as much as 20%, depending on the degree of cigarette use.